Class Imbalance Mitigation for TCGA-UT

Abstract

Class imbalance in computational pathology depends on the input regime: labeled image patches support ordinary classification, whereas whole-slide image (WSI) diagnosis is commonly modeled as weakly supervised learning over patch-feature bags. This report provides an overview and controlled comparison of representative imbalance-mitigation method classes on TCGA-UT, a typical long-tailed histopathology dataset. Two native-regime benchmarks are evaluated on shared slide-disjoint splits and three seeds. On the patch benchmark with frozen Virchow2 embeddings, center-focused affinity loss (CFAL) reaches the highest mean macro F1 (0.807), clearly above plain cross-entropy (0.776) and Scholz-style soft-metric hybrids (0.782), whereas ProGAN augmentation and focal loss do not. On the WSI-bag benchmark, RankMix consistently improves over MIL CE on balanced accuracy and calibration across all three seeds and has the highest mean balanced accuracy under the fixed protocol, while MDE-MIL—a dual-expert ensemble without multimodal distillation—has the highest mean macro F1 but with wider seed spread and overlapping uncertainty bands across methods. The report therefore frames imbalance mitigation as a regime-dependent design choice rather than a universally transferable recipe.

1 Introduction

Class imbalance changes what a pathology classifier is rewarded for learning. When common tumor types dominate training data, high aggregate accuracy can coexist with unreliable rare-class recall, especially if evaluation emphasizes prevalence-sensitive summaries. Long-tailed learning and medical-image evaluation literature therefore motivate classwise recall, macro F1, balanced accuracy, and calibration-aware summaries as primary outcomes in imbalanced settings (Saini and Susan, 2023; Zhang et al., 2023; Mosquera et al., 2024).

A common source of ambiguity in pathology imbalance studies is the input regime. Some methods are naturally image-level methods: they assume labeled images, image embeddings, or directly generated image samples. Other methods are naturally WSI-level methods: they assume weak labels over bags of instances and use attention, instance ranking, or bag-level representation learning. Comparing such methods in one pooled leaderboard can confound the imbalance mechanism with changes in supervision, input size, backbone, and aggregation.

TCGA-UT is useful for separating these questions because it supports two legitimate tasks with different native inputs. The dataset provides labeled histopathology patches cropped from pathologist-selected tumor regions and slide identities that can be grouped into WSI-level bags. Patch labels are class labels, not dense pixel masks or cell-level annotations. This report therefore uses the available supervision directly: patch-level methods are evaluated on controlled patch classification with frozen Virchow2 embeddings, while MIL methods are evaluated on slide-level feature bags built from the same foundation-model family.

The purpose is not to introduce a new mitigation method or to reproduce every cited method in all implementation detail. Instead, the main contribution is a benchmarking report that maps major mitigation classes onto a typical histopathology long-tail setting and compares their behavior under a shared protocol. The report makes three contributions: it separates patch-level and WSI-bag imbalance evaluation rather than mixing them in one leaderboard; it implements representative data-level, loss-level, hybrid, synthetic-generation, and MIL-specific interventions under fixed splits and frozen Virchow2 features; and it reports discrimination, classwise, calibration, and validation-tuning evidence so that weak fixed-protocol results can be interpreted as either robust failures or possible under-tuning. Method fidelity is defined by implementing the defining imbalance mechanism in the regime for which that mechanism was designed, while holding the backbone family, split, input budget, and evaluation code fixed within each benchmark.

2 Dataset

TCGA-UT is derived from diagnostic *whole-slide images* (WSIs) in The Cancer Genome Atlas (TCGA), a pan-cancer resource that links molecular and clinical cancer data with large image collections (Tomczak et al., 2015). The TCGA-UT release turns this slide archive into a computational-pathology image dataset by extracting hematoxylin and eosin (H&E) tumor *patches* from pathologist-selected *uniform tumor regions*. It contains 1,608,060 RGB patches of size 256×256 pixels from 8,736 diagnostic slides and 32 *cancer-type labels*. The patch labels are slide- or cancer-type labels, not dense tissue masks or cell-level annotations (Komura et al., 2022).

This structure makes TCGA-UT well suited for an imbalance study because the imbalance is native to the dataset rather than imposed only after sampling. The largest class is glioblastoma multiforme (GBM; 792 slides), while the smallest class is diffuse large B-cell lymphoma (DLBC; 28 slides). The resulting slide-level imbalance ratio is 28.3. As a consequence, the dataset exposes the practical tension that motivates this study: methods must learn from many visually related cancer categories while preserving performance on classes with far fewer slide-level examples.

Property	Value
Cancer-type labels	32
Diagnostic slides	8,736
TCGA-UT patches	1,608,060 patches of 256×256 pixels
Largest class	Glioblastoma multiforme (GBM), 792 slides
Smallest class	Diffuse large B-cell lymphoma (DLBC), 28 slides
Slide-level imbalance ratio	28.3
Shared split size per seed	6,116 train, 1,310 validation, 1,310 test slides
Controlled patch input	Resolution level 0; up to 30 deterministic patches per slide; one frozen Virchow2 embedding per patch (Section 2.2)
WSI-bag input	Up to 30 deterministic frozen Virchow2 feature instances from the same slide identities

Table 1: Dataset summary used by the shared patch and WSI-bag benchmarks.

2.1 Data Units and Terminology

Several data units are involved in the benchmark, and the distinction matters for method comparison. A *slide* or *WSI* is one diagnostic histopathology image associated with one cancer-type label. A *tumor region* is a pathologist-selected area of a slide from which TCGA-UT crops groups of patches. A *patch* is one 256×256 RGB crop selected by the controlled manifest. A *patch embedding* is the frozen Virchow2 vector computed from that crop and consumed by the patch-level linear classifier. A *feature chunk* is a saved frozen-feature representation associated with a slide or slide subset. A *WSI bag* is the collection of frozen patch-feature instances used by the attention-MIL model for one slide-level prediction. *Head*, *body*, and *tail* classes refer to fixed support tiers derived from the full TCGA-UT slide distribution when reporting aggregate class-group metrics on the held-out test split. A *slide-disjoint split* means that all patches and

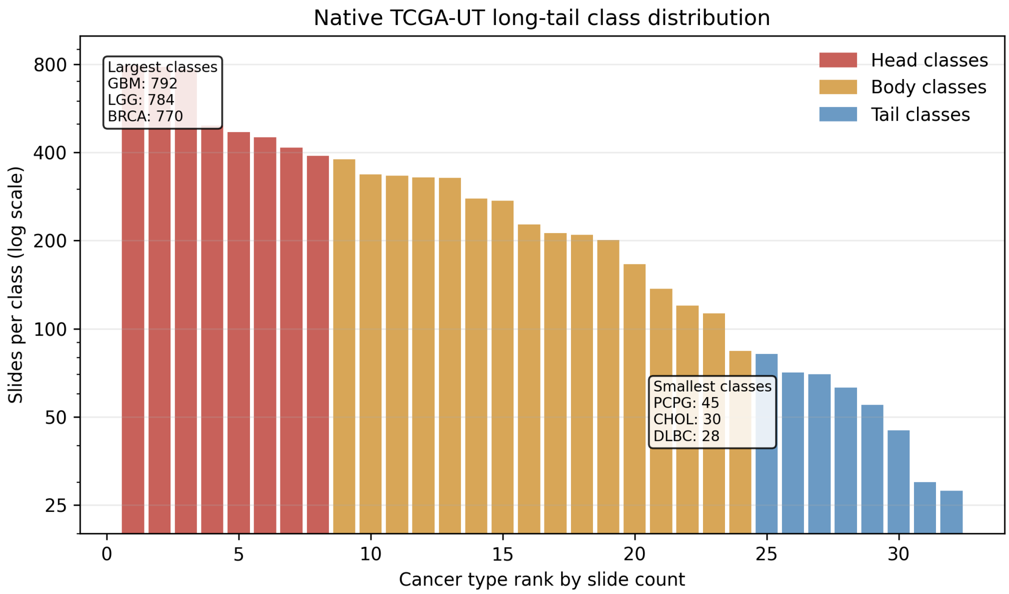
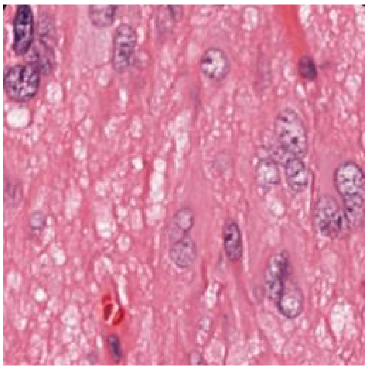


Figure 1: Native TCGA-UT slide support by cancer-type rank. The long-tailed distribution motivates metrics that weight classes more evenly than aggregate accuracy.

features from one slide stay in exactly one of train, validation, or test. Table 12 in Appendix A lists the same terms in compact form for quick reference.



Field	Example value
Cancer type	Glioblastoma multiforme
Resolution level	0
Patch size	256×256 RGB pixels
Source unit	One crop from a selected tumor region
Patch role	Image-level sample if selected by the controlled manifest
WSI-bag role	Candidate instance in the slide's feature bag

Table 2: Example TCGA-UT data point copied from the shared raw dataset. The image is a single patch, while the table shows how that patch relates to the manifest fields used by the benchmark.

2.2 Shared Splits and Benchmark Inputs

For each seed, slides are split into 6,116 training, 1,310 validation, and 1,310 test slides. The same slide-level assignments are reused by both benchmarks so that no patches from the same slide can appear in different splits. This is the main leakage-control mechanism: the patch benchmark and the WSI-bag benchmark see different input units, but they inherit the same train/validation/test slide partition.

The patch benchmark derives a deterministic controlled patch manifest from the shared slide split. It uses resolution level 0 and samples up to 30 patches per slide with a fixed rule before method-specific training begins. The cap is a compromise between data coverage and slide-level control: it uses the empirical median number of raw JPG patches per slide at resolution level 0 (median 30, mean 31.1, range 10–80) while preventing slides with many extracted tumor-region patches from dominating the patch-level loss. At this resolution, 8,686 of 8,736 slides have at least 30 raw patches, so the cap retains the typical per-slide patch evidence while limiting only the larger slide folders. Each selected patch is encoded once with a frozen Virchow2 model, and all patch methods train the same lightweight classifier head on these embeddings (Zimmermann et al., 2024; Vorontsov et al., 2024). The encoder output is concatenated as $\mathrm{concat}(\mathrm{CLS}, \mathrm{mean}(\text{patch tokens}))$, yielding 2,560-dimensional float16 vectors that are cast to float32 before classifier training. This design isolates imbalance mechanisms from both patch-selection policy and end-to-end image-backbone learning.

The WSI-bag benchmark applies the same per-slide evidence budget: when a frozen feature bag contains more than 30 instances, it is deterministically subsampled to 30 evenly spaced instances before attention-MIL training and evaluation. This keeps compute bounded and avoids a systematic advantage for slides with larger extracted feature bags. The two benchmarks still differ in supervision unit and aggregation: patch models classify each selected crop independently, whereas WSI-bag models predict one slide label from an attention-weighted set of up to 30 instances.

3 Methods

The methods are organized by the level at which they intervene in the imbalanced learning problem. Cross-entropy (CE) is treated as the no-mitigation reference condition: it uses the observed training distribution as given and does not rebalance samples, labels, costs, representations, or generated data. The remaining methods are tested imbalance interventions grouped according to the taxonomy used in the class-imbalance literature: data-level sampling and augmentation, algorithm-level losses, hybrid sampling–loss methods, synthetic generation, and representation or architecture-level methods (Saini and Susan, 2023; Zhang et al., 2023).

The comparison is regime-specific. Patch methods consume frozen Virchow2 embeddings of controlled patches and optimize one shared MLP classifier head (linear for most methods). WSI-bag methods consume capped frozen Virchow2 feature bags and optimize one shared attention-MIL head. Both benchmarks therefore rely on the same pathology foundation-model family, but they differ in supervision unit: one embedding per selected patch versus aggregation over a bounded set of instances on a slide. Method differences within a benchmark therefore correspond to imbalance mechanisms rather than changes in split, evaluation code, or encoder choice.

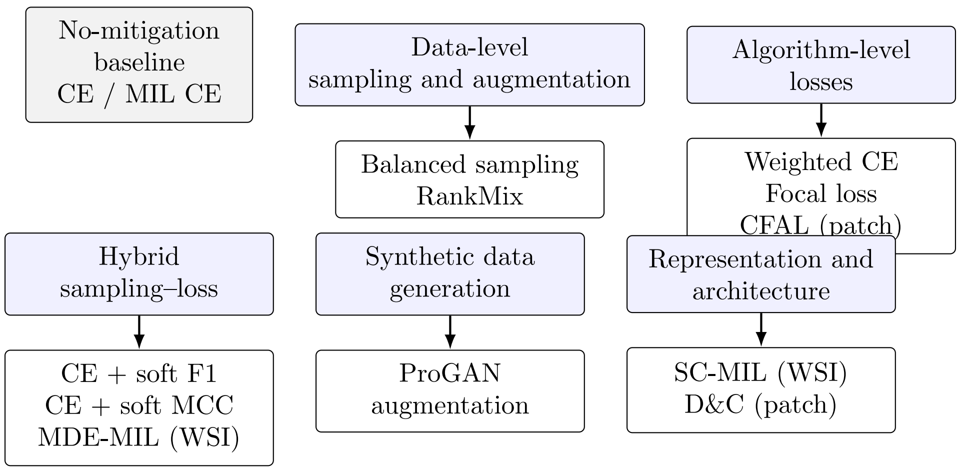


Figure 2: Method organization used in the benchmark. Cross-entropy is the no-mitigation reference, while the tested alternatives intervene through sampling, loss design, synthetic data, WSI feature augmentation, or representation learning.

3.1 Notation

Let C be the number of cancer-type classes and let n_c denote the number of training examples from class c. For a labeled patch or slide-level bag with label $y \in \{1, \ldots, C\}$, the model outputs logits $\mathbf{z} \in \mathbb{R}^C$ and class probabilities

$$p_c = \frac{\exp(z_c)}{\sum_{j=1}^{C} \exp(z_j)}. \tag{1}$$

For WSI-bag methods, a slide is represented as a bag $B = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ of frozen patch-feature instances. The attention-MIL backbone maps each bag to a bag embedding $\mathbf{b}(B)$ and class logits. Unless a method explicitly changes the sampler, objective, or representation loss, the empirical class frequencies in the training set define the optimization signal.

3.2 Cross-Entropy Baseline

Unweighted cross-entropy is the baseline because it is the standard supervised objective without an explicit class-imbalance correction. For an example with label y, the loss is

$$\mathcal{L}_{\text{CE}}(\mathbf{z}, y) = -\log p_y. \tag{2}$$

In the patch benchmark, this objective trains the shared linear head on frozen Virchow2 patch embeddings from the controlled manifest. In the WSI-bag benchmark, the same objective trains the shared attention-MIL model on slide-level feature bags. In both cases, CE uses the observed class distribution directly: majority classes appear more often in minibatches and therefore contribute more loss terms over an epoch. This makes CE an appropriate reference for asking whether an imbalance-specific intervention actually improves the rare-class tradeoff.

3.3 Data-Level Sampling and Augmentation

Data-level methods change what the optimizer sees, rather than changing the mathematical penalty assigned to one observed example. In the taxonomy, this family includes resampling, augmentation, and WSI-specific feature mixing strategies that make minority or informative examples more visible during training (Saini and Susan, 2023).

3.3.1 Balanced Sampling

Balanced sampling changes minibatch composition by sampling training examples with probabilities inversely related to their class support. For an example i with class y_i, the sampler weight is proportional to

$$q_i \propto \frac{1}{n_{y_i}}. \tag{3}$$

After normalization across the training set, this makes the expected class exposure closer to uniform even though the underlying dataset remains long-tailed. The patch variant applies this sampler to patch examples and still optimizes CE. The MIL variant applies the same idea at slide-bag level and also optimizes CE. Balanced sampling therefore tests a pure data-level intervention: it asks whether equalizing training exposure is enough without changing the loss formula.

3.3.2 RankMix

RankMix is a WSI-native data-augmentation method for weakly supervised slide classification. Ordinary image mixup assumes two inputs with compatible spatial shapes, but WSI bags can contain different numbers of feature instances and lack instance-level labels. RankMix addresses this by mixing representative feature subsequences rather than raw slides (Chen and Lu, 2023).

The tested implementation follows the defining two-stage teacher–student procedure. First, a teacher attention-MIL model is trained with slide labels for 30 epochs using the same MIL architecture and optimization settings as plain MIL CE; when available, the matched plain MIL CE checkpoint for the same seed is reused as the teacher. For each bag $B = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ and target class, the teacher assigns ranking scores to instances using its softmax probability for that class; the top-k instances are kept as a representative subsequence, where $k = \min(|B_a|, |B_b|)$ for the two bags being mixed, and their original within-bag order is preserved after selection. The student MIL model is then reinitialized and trained on bags built from these ranked subsequences.

Mixing itself follows the mixup idea in feature space. For two bags a and b, let H_a and H_b denote the selected instance-feature matrices and $\mathbf{y}_a, \mathbf{y}_b$ the corresponding one-hot slide labels. A scalar mixing coefficient $\lambda \in (0, 1)$ controls how much of each bag is retained:

$$\widetilde{H} = \lambda H_a + (1 - \lambda) H_b, \qquad \widetilde{\mathbf{y}} = \lambda \mathbf{y}_a + (1 - \lambda) \mathbf{y}_b. \tag{4}$$

When λ is close to one, the mixed bag resembles bag a; when it is close to zero, it resembles bag b. Values near $\lambda = \frac{1}{2}$ interpolate more evenly between the two sources. RankMix does not fix λ to a single value. Instead, each mixed example draws

$$\lambda \sim \mathrm{Beta}(\alpha, \alpha), \tag{5}$$

independently. The Beta distribution is a standard choice on the open interval $(0, 1)$ because it is flexible and symmetric when both shape parameters are equal. With density

$$f(\lambda; \alpha, \alpha) \propto \lambda^{\alpha-1}(1 - \lambda)^{\alpha-1}, \qquad \lambda \in (0, 1), \tag{6}$$

the parameter $\alpha > 0$ controls how strongly mixing concentrates around the endpoints versus the center. For $\alpha = 1$, $\mathrm{Beta}(1, 1)$ is the uniform distribution on $(0, 1)$, so every mixing weight is equally likely and the procedure reduces to ordinary symmetric mixup. Larger α pushes λ toward $\frac{1}{2}$, producing more balanced interpolations; smaller α pushes mass toward 0 and 1, producing mixtures that stay closer to one source bag. The benchmark fixes $\alpha = 1.0$, matching the default used in the RankMix reference implementation. The student is trained with soft-label cross-entropy on the mixed bags and labels $\widetilde{\mathbf{y}}$. This mechanism is categorized as data-level augmentation because it increases the diversity of the training bags in feature space while retaining the same attention-MIL prediction architecture.

3.4 Algorithm-Level Losses

Algorithm-level methods keep the training examples fixed but alter how errors are penalized. They are cost-sensitive or difficulty-sensitive objectives: minority examples may receive larger explicit weights, or easy examples may contribute less to the gradient (Saini and Susan, 2023; Scholz et al., 2024).

3.4.1 Class-Weighted Cross-Entropy

Class-weighted CE multiplies each example's CE term by a class-dependent cost:

$$\mathcal{L}_{\mathrm{WCE}}(\mathbf{z}, y) = -w_y \log p_y, \qquad w_c \propto \frac{1}{n_c}. \tag{7}$$

The implementation normalizes the inverse-frequency weights so that the average scale of the loss remains comparable across methods. This method leaves the sample order and model architecture unchanged, but increases the loss contribution of classes with fewer training examples. It is evaluated for both patch classification and attention-MIL.

3.4.2 Focal Loss

Focal loss reduces the contribution of confidently classified examples and concentrates optimization on examples that remain difficult. With focusing parameter γ, the multiclass focal objective is

$$\mathcal{L}_{\text{focal}}(\mathbf{z}, y) = -(1 - p_y)^{\gamma} \log p_y. \tag{8}$$

This study uses $\gamma = 2.0$ for the patch and WSI-bag variants. The method is relevant to imbalance because many minority-class examples can remain hard under a classifier dominated by head-class gradients, but the focal term does not directly encode the class counts. It is therefore a difficulty-sensitive loss rather than a sampler or explicit prior correction.

3.4.3 Center-Focused Affinity Loss (CFAL)

Center-focused affinity loss (CFAL) replaces the linear softmax head with learnable class prototypes and an affinity-based training objective (Mahbub et al., 2024). The motivation is that long-tailed histology classes can remain visually similar in embedding space even when a frozen encoder already separates them coarsely; pulling examples toward compact class centers and penalizing confusion with nearby wrong-class prototypes can therefore improve tail-class boundaries without changing the sampler.

The tested patch implementation keeps the shared MLP trunk that maps a frozen Virchow2 embedding $\mathbf{x}$ to a hidden representation $\mathbf{z} = f(\mathbf{x}) \in \mathbb{R}^{512}$, but replaces the final linear map with one learnable prototype $\mathbf{w}_c \in \mathbb{R}^{512}$ per class. Both embeddings and prototypes are L2-normalized before affinity computation. The Gaussian affinity between example i and class c is

$$a_{ic} = \exp\left(-\frac{\|\hat{\mathbf{z}}_i - \hat{\mathbf{w}}_c\|_2^2}{\sigma}\right), \qquad \hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|_2}, \tag{9}$$

where σ controls how quickly affinity decays with prototype distance. At inference, class scores are $\log a_{ic}$ followed by softmax normalization. The affinities $a_{ic} \in (0, 1]$ are geometric similarity scores, not class probabilities: they are trained with a margin objective and never optimized to satisfy likelihood or calibration constraints. Applying softmax to $\log a_{ic}$ therefore yields a convenient ranking score for argmax decisions, but it should not be read as a calibrated estimate of $P(y = c \mid \mathbf{x})$ without additional post-hoc mapping.

Training optimizes a margin term on affinities rather than cross-entropy on logits. For batch index i with label y_i, define the per-example margin loss

$$\ell_i = \sum_{c \neq y_i} \max\bigl(0,\, \lambda + a_{ic} - a_{i,y_i}\bigr), \tag{10}$$

which penalizes wrong-class affinities that exceed the true-class affinity by less than margin λ. A diversity regularizer $R(\mathbf{W})$ encourages prototypes to spread apart by penalizing deviations of pairwise squared distances from their mean. The total objective combines class reweighting, a center-focused local weight, and prototype diversity:

$$\mathcal{L}_{\text{CFAL}} = \frac{1}{B} \sum_{i=1}^{B} \frac{1}{E_{y_i}} (1 - a_{i,y_i})^{\gamma}\, \ell_i + R(\mathbf{W}), \tag{11}$$

where E_c is the effective number of training examples in class c derived from class count n_c and hyperparameter β, and γ up-weights examples whose true-class affinity remains low. Unlike the Scholz-style hybrids, CFAL does not use balanced oversampling: it is categorized as an algorithm-level patch method because it changes the head and loss while keeping the natural training distribution.

The fixed-protocol benchmark uses $\lambda = 0.1$, $\beta = 0.999$, and $\gamma = 2.0$ following Mahbub et al.; Section 5.4 additionally sweeps $\gamma \in \{0.5, 1, 2, 5\}$ on validation macro F1. Two implementation adaptations are required for frozen Virchow2 embeddings. First, the paper reports $\sigma = 130$ for AlexNet FC features; on 2,560-dimensional Virchow2 vectors that scale collapses affinities unless embeddings are normalized, so this benchmark uses L2 normalization with $\sigma = 1.0$. Second, CFAL is evaluated only in the patch regime because the source method assumes instance-level labels rather than weak slide-level bags.

3.5 Hybrid Sampling–Loss Methods

Hybrid methods combine data-level and algorithm-level changes. The Scholz-style representatives are placed here because the tested variants use class-balanced oversampling together with metric-derived losses. This coupling matters: their effect cannot be attributed only to a new loss or only to a new sampler.

3.5.1 Scholz-Style CE + Soft F1

Scholz et al. motivate losses derived from metrics that are already used to evaluate imbalanced medical classification, including F1 and Matthews correlation coefficient (MCC) (Scholz et al., 2024). The intuition is that CE optimizes the log-probability of the correct class one example at a time, whereas macro F1 evaluates a different object: the balance between precision and recall at class level. In a long-tailed dataset, this distinction matters because a model can reduce CE substantially by fitting frequent classes while still leaving tail-class recall or precision weak.

The obstacle is that ordinary F1 is computed from hard counts after prediction, so it is not directly usable as a gradient-based training loss. The Scholz-style soft-F1 objective replaces those hard counts with probability-weighted counts inside each minibatch. For one class, a confident correct prediction contributes almost one soft true positive, a confident prediction for the wrong class contributes to soft false positives, and low probability on a true class contributes to soft false negatives. Formally, for one-hot labels y_i and predicted probabilities $\hat{y}_i$:

$$\mathrm{TP} = \sum_i y_i \hat{y}_i, \quad \mathrm{FP} = \sum_i (1 - y_i)\hat{y}_i, \quad \mathrm{FN} = \sum_i y_i (1 - \hat{y}_i). \tag{12}$$

These terms keep the familiar meaning of a confusion matrix but make every component differentiable with respect to the model probabilities. They then define soft precision, soft recall, and soft F1:

$$\mathrm{F1}_{\mathrm{soft}} = \frac{2\,\mathrm{precision}_{\mathrm{soft}}\,\mathrm{recall}_{\mathrm{soft}}}{\mathrm{precision}_{\mathrm{soft}} + \mathrm{recall}_{\mathrm{soft}}}. \tag{13}$$

The multiclass implementation applies this construction separately to each class and averages the resulting classwise terms. This macro-style averaging is the imbalance-aware part: each class contributes one term regardless of how many samples it has in the full dataset. The tested objective does not replace CE completely; it adds the metric-derived term to CE so that optimization still rewards calibrated class probabilities while also pushing the minibatch toward higher macro soft F1:

$$\mathcal{L}_{\mathrm{CE+F1}} = \mathcal{L}_{\mathrm{CE}} + \left(1 - \frac{1}{C}\sum_{c=1}^{C} \mathrm{F1}_{\mathrm{soft},c}\right). \tag{14}$$

The benchmark trains this loss with balanced sampling on the patch manifest. That design is important for interpretation: the loss tries to optimize a macro-style class metric, while the sampler ensures that tail classes occur often enough in minibatches for their soft-F1 terms to provide a regular training signal.

3.5.2 Scholz-Style CE + Soft MCC

The soft-MCC variant uses the same principle as soft F1—replace hard evaluation counts with differentiable probability masses—but starts from the Matthews correlation coefficient (MCC).

Whereas F1 for one class depends mainly on that class's positive predictions, MCC is built from the full confusion matrix and therefore reacts to false positives, false negatives, and true negatives at the same time. In binary problems it is often preferred in imbalanced settings because it can remain informative when one class dominates the dataset: accuracy can look high while predictions are still systematically biased, but MCC penalizes both kinds of mistakes jointly (Scholz et al., 2024).

Binary MCC as a correlation coefficient. For one class treated as positive, write TP, TN, FP, and FN for the four confusion-matrix cells. The standard MCC is

$$\mathrm{MCC} = \frac{\mathrm{TP}\cdot\mathrm{TN} - \mathrm{FP}\cdot\mathrm{FN}}{\sqrt{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})}}. \tag{15}$$

The numerator compares co-occurrence of correct positive and correct negative decisions with co-occurrence of the two error types; the denominator scales by how many examples fall into each margin of the confusion matrix. MCC equals +1 for perfect predictions, 0 for chance-level association, and can become negative when predictions are worse than random. This makes it a correlation-style score rather than a prevalence-weighted rate.

Multiclass MCC in a minibatch. For C cancer types, Scholz et al. extend the idea to a minibatch of size N by treating labels and predictions as C-dimensional vectors per example (Scholz et al., 2024). Let $Y \in \{0,1\}^{N\times C}$ be the one-hot label matrix and $\widehat{Y} \in [0,1]^{N\times C}$ the corresponding softmax probabilities, with rows $\mathbf{y}_i$ and $\hat{\mathbf{y}}_i$. Three aggregate quantities describe how mass is distributed inside the batch:

$$s = \sum_{i=1}^{N}\sum_{c=1}^{C} Y_{ic}\widehat{Y}_{ic}, \qquad t_c = \sum_{i=1}^{N} Y_{ic}, \qquad p_c = \sum_{i=1}^{N} \widehat{Y}_{ic}. \tag{16}$$

Here s is the total probability-weighted agreement between labels and predictions: each example contributes $\hat{y}_{i,y_i}$, the predicted mass on its true class. The terms t_c and p_c are the batch totals of true and predicted mass for class c. They play the role of marginals in a confusion summary. If predictions collapse onto a few head classes, the vector $(p_1,\ldots,p_C)$ becomes skewed even when argmax accuracy looks acceptable; MCC is sensitive to that mismatch because it compares the joint alignment s with what would be expected from the marginals alone.

The multiclass MCC used in training can be written as a normalized covariance between the label matrix and the prediction matrix:

$$\mathrm{MCC}_{\mathrm{mc}} = \frac{Ns - \sum_{c=1}^{C} t_c p_c}{\sqrt{N^2 - \sum_{c=1}^{C} p_c^2}\,\sqrt{N^2 - \sum_{c=1}^{C} t_c^2}}. \tag{17}$$

The numerator $Ns - \sum_c t_c p_c$ increases when true and predicted mass line up example by example and decreases when correct predictions are no better than what the marginal totals would suggest by chance. The two square-root terms measure how spread out the batch-level label totals and prediction totals are across classes; they prevent the score from exploding when one minibatch happens to contain only a few classes. Equation (17) reduces to the familiar binary form in (15) when $C = 2$.

Soft MCC loss. Hard MCC replaces $\widehat{Y}$ by one-hot argmax predictions, which again blocks gradient-based training. The soft-MCC objective keeps the structure of (17) but plugs in softmax probabilities directly:

$$\mathrm{MCC}_{\mathrm{soft}} = \frac{Ns - \sum_{c=1}^{C} t_c p_c}{\sqrt{N^2 - \sum_{c=1}^{C} p_c^2}\,\sqrt{N^2 - \sum_{c=1}^{C} t_c^2}}. \tag{18}$$

Every term in (16) and (18) is differentiable with respect to the logits that produce $\widehat{Y}$. A confident correct prediction increases s; probability placed on the wrong classes increases some p_c while leaving the matching t_c unchanged, which lowers the numerator; and overly peaked prediction vectors reduce the prediction-side denominator term, limiting the score even if a few examples are classified correctly. As with soft F1, the training objective turns a score to be maximized into a loss to be minimized and adds it to CE:

$$\mathcal{L}_{\mathrm{CE+MCC}} = \mathcal{L}_{\mathrm{CE}} + (1 - \mathrm{MCC}_{\mathrm{soft}})\,. \tag{19}$$

This variant is also evaluated only in the patch benchmark and is trained with class-balanced oversampling. The soft-F1 and soft-MCC variants therefore ask a focused question: once tail classes are regularly presented to the optimizer, does adding a differentiable version of a class-balanced evaluation metric improve the controlled patch classifier beyond CE and simple balanced sampling?

3.6 Synthetic Data Generation

Synthetic generation is data-level in spirit but is separated because it introduces new generated samples rather than resampling observed ones. In histopathology, this family is attractive because deleting majority-class tissue wastes data, while naive image-space interpolation often produces unrealistic pathology images (Saini and Susan, 2023; Ruiz-Casado et al., 2024). The practical hope is simple: if real minority-class patches are scarce, a generator may learn enough of their stain, texture, and tissue morphology distribution to provide additional plausible training examples.

3.6.1 ProGAN Augmentation

The ProGAN experiment adapts the GAN-based histopathology augmentation idea of Ruiz-Casado et al. to multiclass TCGA-UT patches (Ruiz-Casado et al., 2024). A generative adversarial network (GAN) has two coupled models. The generator G maps random latent vectors to synthetic images, while the discriminator D tries to distinguish real training patches from generated patches. Training alternates between making D better at this distinction and making G produce images that D cannot reliably reject. The standard objective expresses this competition as

$$\min_G \max_D \mathbb{E}_{\mathbf{x}\sim p_{\mathrm{data}}}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{u}\sim p(\mathbf{u})}[\log(1 - D(G(\mathbf{u})))]. \tag{20}$$

This equation is not a classification loss for the final TCGA-UT classifier. It is the pretraining objective for producing additional patch images. After generation, the classifier simply sees a larger patch training set.

ProGAN modifies ordinary GAN training by growing the image resolution gradually. Instead of asking the generator to synthesize 256×256 histology patches from the start, training begins at a coarse resolution where global color and texture structure is easier to model. Higher-resolution layers are then introduced step by step until the target patch size is reached. This staged procedure is intended to stabilize synthesis and reduce the risk that the generator learns only a narrow set of repeated images. In this benchmark, one class-specific generator is trained for every training class whose patch count is below the head-class training count. Progressive training uses seven resolution stages from 4×4 to 256×256 pixels, with 25 epochs per stage, a fade-in fraction of 0.5 for newly added layers, latent dimension 128, base channel width 512, Adam optimization with learning rate 2×10^{-4} and $\beta_1 = 0.5$, and the depth-dependent batch schedule of Ruiz-Casado et al. When a class has more than 2,048 real training patches, at most 2,048 are drawn uniformly at random with a deterministic per-class seed for ProGAN training only; the downstream classifier still uses the full real training set together with the generated patches. Synthetic patch counts are set to the head-class training count minus the class training count, so each augmented tail class matches the largest real training class. Inception FID is computed for every augmented class between the generated images and the real training subset used for that class-specific generator; seed 0 has median FID 178.5, with values ranging from 103.0 to 315.9. No generated samples are filtered by FID or other quality thresholds. Real and synthetic RGB patches are embedded with the same frozen Virchow2 encoder before classifier training, so the augmentation is evaluated in the same embedding regime as the other patch methods. Synthetic images are used only as extra patch-classification training examples; they are not converted into synthetic WSI bags.

3.7 Representation and Architecture-Level Methods

Representation-level methods try to make rare classes more separable in feature space rather than only changing sampling or per-example loss weights. This family is especially relevant for computational pathology, where WSI labels supervise bags of many unlabeled instances and class imbalance can occur both across slides and within a positive slide (Juyal et al., 2024).

3.7.1 Divide-and-Conquer (D&C)

Divide-and-conquer patch classification decomposes a long-tailed multiclass problem into three binary subproblems, trains separate experts on cluster-sampled subsets, and fuses their representations for the final decision (Nouyed et al., 2024). Nouyed et al. originally partition HiPath patches into tumor versus benign-in-cancerous-slide versus benign-from-benign-slide sets and train experts on *A vs B*, *A vs C*, and *A vs (B+C)* with k-means cluster sampling and z-score stratification so the majority side matches the minority count while preserving texture diversity. Because TCGA-UT patches have cancer-type labels but no corresponding benign-region groups, the valid adaptation fixes three mutually exclusive groups before model fitting from the dataset-level slide-support tiers defined in Section 3.8: set A is the eight tail classes, set B is the sixteen body classes, and set C is the eight head classes. This partition is independent of validation and test predictions and prevents any class from appearing on both sides of a binary subproblem.

Each expert uses the shared 2,560→512 MLP trunk and a binary linear head. For epochs 1–20, the three experts are trained independently on cluster-sampled binary datasets with BCE; for epochs 21–30, the expert trunks are frozen and a fusion head is trained on the full natural training manifest with multiclass CE. Fusion follows an M2-lite design: concatenate the three 512-d expert embeddings, apply dropout, and map to 32 classes with a linear layer rather than the paper's PCA/RandomForest fusion modes. Cluster sampling uses $k = 10$ clusters, five z-score bins, and a 20,480-patch cap for k-means fitting; Section 5.4 additionally sweeps $k \in \{5, 10, 15, 20\}$ on validation macro F1. D&C is evaluated only in the patch benchmark because the source method assumes instance-level labels rather than weak slide-level bags.

3.7.2 SC-MIL

SC-MIL extends supervised contrastive learning to MIL by applying the contrastive objective to bag representations instead of assigning noisy slide labels to individual patches (Juyal et al., 2024). The motivation is that a slide label usually does not describe every patch in a bag. A cancer-type label can be correct for the slide while many instances contain background, normal tissue, artifacts, or weakly informative morphology. Applying contrastive learning directly to instances would therefore treat many patches as positives or negatives using labels they do not truly carry. SC-MIL avoids that mismatch by contrasting the aggregated slide-level representation.

The attention-MIL backbone first maps bag B_i to an embedding $\mathbf{b}_i = \mathbf{b}(B_i)$. A projection head g then produces normalized contrastive features $\mathbf{r}_i = g(\mathbf{b}_i)$. In the contrastive view, each bag in a minibatch is an anchor. Bags with the same cancer-type label are positives because their slide-level representations should become closer, while bags from other classes act as negatives because their representations should remain separated. For anchor i, bags of the same class form the positive set $P(i)$, and the supervised contrastive term is

$$\mathcal{L}_{\mathrm{SCL}} = -\frac{1}{\sum_i |P(i)|} \sum_i \sum_{p \in P(i)} \log \frac{\exp(\mathbf{r}_i^\top \mathbf{r}_p / \tau)}{\sum_{a \neq i} \exp(\mathbf{r}_i^\top \mathbf{r}_a / \tau)}. \tag{21}$$

The numerator rewards similarity to same-class bags. The denominator compares that similarity against all other bags in the minibatch, so the loss favors representations where class clusters are compact and mutually separated. The temperature τ controls how sharply these similarities are weighted; this benchmark uses $\tau = 1.0$.

SC-MIL does not rely on contrastive learning alone. The contrastive term shapes the representation space, while CE trains the final class predictor. The tested variant combines them through a linear curriculum:

$$\mathcal{L}_t = \beta_t \mathcal{L}_{\mathrm{SCL}} + (1 - \beta_t)\mathcal{L}_{\mathrm{CE}}, \qquad \beta_t = 1 - \mathrm{progress}_t. \tag{22}$$

Early training therefore emphasizes class-structured bag embeddings, when the classifier head is still unstable. As training progresses, β_t decreases and more weight moves to CE, so the learned representation is tied back to the actual slide-level classification objective. Activation diagnostics count positive pairs during training to confirm that the contrastive component is active.

3.7.3 MDE-MIL

MDE-MIL addresses long-tailed WSI classification with distribution-specific experts and a cross-expert consistency term (Ling et al., 2025). The full method combines a dual-expert ensemble with multimodal distillation from a pathology foundation model; this benchmark implements the *ensemble branch only*: a shared attention aggregator, two expert classification heads, and a logit consistency loss. Multimodal CONCH distillation is excluded so that the effect of dual-distribution training can be isolated under the same frozen Virchow2 bags as the other MIL methods.

The shared trunk matches plain attention MIL: each bag $B = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ is encoded instance-wise, attention weights are computed with a linear scorer and softmax over instances, and the weighted sum yields a bag embedding $S(B) \in \mathbb{R}^d$. Two separate feed-forward experts then map S to logits:

$$E_U(S) = W_U\, \phi_U(S) + b_U, \qquad E_B(S) = W_B\, \phi_B(S) + b_B, \tag{23}$$

where ϕ_U and ϕ_B are two-layer MLPs with ReLU and dropout. Expert U is trained on the natural slide distribution and expert B on a class-balanced slide distribution.

Each training step draws one minibatch from each distribution. The unbalanced loader shuffles slide bags as in MIL CE; the balanced loader uses the same inverse-frequency weights as balanced-sampling MIL. For batches (B_U, y_U) and (B_B, y_B), the classification loss is

$$\mathcal{L}_{\mathrm{cls}} = \mathrm{CE}\big(E_U(S(B_U)), y_U\big) + \mathrm{CE}\big(E_B(S(B_B)), y_B\big), \tag{24}$$

and the cross-expert consistency term penalizes disagreement on the *same* bag embedding:

$$\mathcal{L}_{\mathrm{con}} = \mathrm{MSE}\big(E_U(S(B_U)), E_B(S(B_U))\big) + \mathrm{MSE}\big(E_B(S(B_B)), E_U(S(B_B))\big). \tag{25}$$

The total objective is $\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{con}}\mathcal{L}_{\text{con}}$. The fixed-protocol benchmark uses $\lambda_{\text{con}} = 0.25$, matching the Camelyon+-LT default reported by Ling et al.; Section 5.4 additionally evaluates $\lambda_{\text{con}} \in \{0.1, 0.25, 0.3\}$ on validation macro F1. Unlike RankMix, MDE-MIL does not use a teacher model or student reinitialization: both experts share the aggregator throughout training.

At validation and test time, each slide bag is evaluated once in natural order with mean-logit ensemble inference:

$$\hat{\mathbf{z}}(B) = \tfrac{1}{2}\big(E_U(S(B)) + E_B(S(B))\big). \tag{26}$$

MDE-MIL is categorized as a hybrid WSI method because it couples two sampling distributions with a consistency regularizer rather than changing only the sampler or only the per-example loss.

3.8 Training Protocol and Reproducibility

All benchmark hyperparameters are fixed in the experiment configuration and are not tuned on the validation split. Validation metrics are written for inspection, but model selection, early stopping, and test reporting all use the weights after the final training epoch. Each benchmark is repeated over seeds 0, 1, and 2, which fix both the slide partition and the PyTorch random seeds used for initialization, shuffling, and sampling.

Table 3 summarizes the shared optimization settings. Patch and WSI-bag classifiers use AdamW with learning rate 10^{-3}, weight decay 10^{-4}, and 30 fixed epochs without early stopping. Patch training uses minibatches of 256 embeddings; WSI-bag training uses minibatches of 32 slide bags. Focal loss uses $\gamma = 2.0$, CFAL uses margin $\lambda = 0.1$, affinity scale $\sigma = 1.0$, center-focused exponent $\gamma = 2.0$, and effective-number $\beta = 0.999$, D&C uses $k = 10$ clusters, five z-score bins, and 20 expert-only epochs before fusion training, RankMix uses $\alpha = 1.0$, SC-MIL uses contrastive temperature $\tau = 1.0$, and MDE-MIL uses consistency weight $\lambda_{\text{con}} = 0.25$ with dual natural and balanced loaders per training step. These method-specific constants follow the cited references or common defaults rather than a validation sweep.

Setting	Patch benchmark	WSI-bag benchmark
Optimizer	AdamW	AdamW
Learning rate	10^{-3}	10^{-3}
Weight decay	10^{-4}	10^{-4}
Epochs	30	30
Early stopping	none	none
Minibatch size	256 patches	32 slide bags
Model selection	final epoch	final epoch
Random seeds	0, 1, 2	0, 1, 2

Table 3: Shared training protocol for both benchmarks. All methods within a benchmark use the same optimizer, schedule, and epoch budget unless noted.

Figure 3 summarizes the two shared classifier heads used in the fixed protocol. Both benchmarks consume frozen 2,560-dimensional Virchow2 features and differ only in how those features are grouped before classification.

The patch head is a two-layer MLP: linear map to 512 units, ReLU, dropout rate 0.1, and a linear output to 32 classes for CE-based methods. CFAL keeps the same trunk but replaces the final linear map with one learnable prototype per class and Gaussian affinities as in (9). The WSI-bag head is an attention-based MIL network on the same feature dimension: each instance passes through the shared encoder, instance weights come from a linear scorer followed by a softmax over instances, the bag representation is their weighted sum, and slide logits are produced by a final linear map. SC-MIL additionally routes bag embeddings through a two-layer projection head of width 256 with ReLU. MDE-MIL keeps the shared encoder and

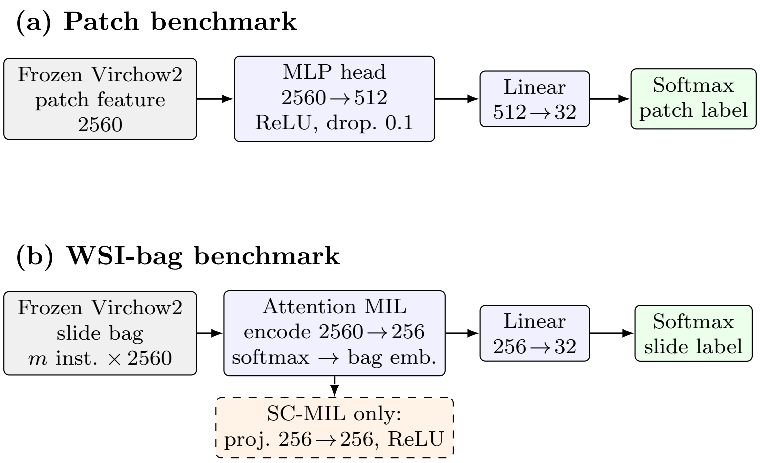


Figure 3: Shared classifier heads for the fixed training protocol. Panel (a) classifies one controlled patch embedding per forward pass. CFAL reuses the MLP trunk but replaces the linear $512 \to 32$ head with learnable prototypes and affinity-based prediction. Panel (b) encodes all instances on a slide, pools them with attention, and predicts one slide-level label; the dashed branch is used only by SC-MIL. MDE-MIL reuses the same attention trunk but replaces the single linear head with two expert heads and mean-logit ensemble inference at evaluation. Virchow2 features are frozen in both benchmarks.

attention pooling but uses two separate expert heads on the bag embedding. The WSI-bag input is capped at 30 instances per slide using deterministic evenly spaced subsampling.

RankMix uses balanced slide-bag sampling during student training. Mixed bags are formed within each minibatch by pairing bags with a random permutation and mixing their teacher-ranked representative subsequences with $\lambda \sim \mathrm{Beta}(\alpha, \alpha)$ and $\alpha = 1.0$ as in (5). The student is reinitialized after teacher preparation and trained for the same 30-epoch budget as the other MIL methods. MDE-MIL instead zips one natural-order batch with one balanced batch at every step, trains both experts on the shared aggregator without a teacher, and evaluates with mean-logit ensembling as in (26).

Head, body, and tail summaries group classes by support in the full TCGA-UT slide distribution rather than by held-out test counts. Cancer types are sorted by dataset slide count; the eight lowest-support classes form the tail tier, the eight highest-support classes form the head tier, and the remaining sixteen classes form the body tier. Tier-wise precision, recall, and F1 on validation and test splits then aggregate only over the predefined classes in each tier.

4 Evaluation

Both benchmarks use seeds 0, 1, and 2. Macro F1 is the primary ranking metric within each benchmark because it weights classes equally and penalizes failure on rare classes. Balanced accuracy, accuracy, classwise precision/recall/F1, and head/body/tail summaries are reported in the main result tables using the tier definitions in Section 3.8. These discrimination metrics answer whether the predicted class is usually correct, but they do not by themselves say whether the reported class probabilities are trustworthy. A model can rank well on macro F1 while assigning near-certainty to many wrong predictions, which is especially relevant in imbalanced medical classification where deployment and thresholding may depend on confidence rather than only on the argmax label (Mosquera et al., 2024). The study therefore reports three complementary probability-quality metrics on held-out softmax outputs; their definitions are given below and their values are summarized in Table 9. Patch and WSI-bag methods are never ranked against each other, because they answer related but different questions with different input units and supervision.

All reported benchmark results use the held-out test split and show mean ± standard deviation over seeds. Table 7 additionally reports paired seed-wise differences for the headline comparisons against each regime's CE baseline; bracketed values list deltas in seed order 0–2 without formal significance testing or bootstrap confidence intervals. Cross-method gaps should therefore be read from seed-consistent paired directions and held-out means; strong comparative wording is reserved for large, seed-stable separations (notably CFAL versus CE on patches), whereas WSI claims emphasize consistent improvements across seeds and highest held-out means when standard deviations overlap. Separate validation summaries are retained as artifacts for model-selection checks. The D&C class partition is fixed from dataset-level slide-support tiers before model fitting and is included in these benchmark comparisons.

4.1 Probability Quality and Calibration

Let N denote the number of held-out examples in one benchmark split, $C = 32$ the number of cancer types, $y_i \in \{1, \ldots, C\}$ the true label, and $\hat{\mathbf{p}}_i \in [0,1]^C$ the corresponding softmax probability vector with $\sum_c \hat{p}_{ic} = 1$. Write $Y_i \in \{0,1\}^C$ for the one-hot encoding of y_i. All three metrics below are computed from the raw model probabilities at the final training epoch, without temperature scaling or other post-hoc calibration.

Negative log likelihood. Cross-entropy training optimizes the log-probability of the correct class. A natural held-out analogue is the average negative log likelihood (NLL), also called multiclass log loss:

$$\mathrm{NLL} = -\frac{1}{N}\sum_{i=1}^{N} \log \hat{p}_{i,y_i}. \tag{27}$$

Lower NLL means the model assigns higher probability mass to the true class on average. NLL therefore measures both whether the top prediction is correct and how sharply probability is placed on that class: a correct but overconfident model and a correct but cautious model can have different NLL even when their hard-error rates match. In implementation, $\hat{p}_{i,y_i}$ is clipped to $[10^{-12}, 1]$ before taking the logarithm so that numerical underflow does not dominate the score when a class receives extremely small probability.

Multiclass Brier score. The Brier score was introduced for probabilistic forecasts and penalizes the entire predicted distribution rather than only the argmax label (Brier, 1950). For multiclass problems it is the mean squared error between $\hat{\mathbf{p}}_i$ and Y_i:

$$\mathrm{BS} = \frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C}\left(\hat{p}_{ic} - \mathbb{1}[y_i = c]\right)^2 = \frac{1}{N}\sum_{i=1}^{N} \|\hat{\mathbf{p}}_i - Y_i\|_2^2. \tag{28}$$

Lower BS is better. The score is minimized when $\hat{\mathbf{p}}_i$ equals the true class indicator and increases when probability leaks into incorrect classes. Unlike NLL, BS is bounded and symmetric in probability errors across classes, so it is less dominated by a single very small $\hat{p}_{i,y_i}$. Together, NLL and BS summarize overall distributional fit on the test split.

Expected calibration error. Discrimination and distributional fit can still leave confidence misaligned with empirical accuracy. Expected calibration error (ECE) measures that gap using confidence bins (Naeini et al., 2015; Guo et al., 2017). For each example define the predicted class $\hat{y}_i = \arg\max_c \hat{p}_{ic}$ and the confidence $c_i = \max_c \hat{p}_{ic}$. Partition examples into $B = 10$ equal-width bins on $[0,1]$ with edges $\tau_0 = 0 < \tau_1 < \cdots < \tau_{10} = 1$ and intervals $(\tau_{b-1}, \tau_b]$. Let

$B_b = \{i : c_i \in (\tau_{b-1}, \tau_b]\}$. The bin accuracy and mean confidence are

$$\mathrm{acc}(B_b) = \frac{1}{|B_b|} \sum_{i \in B_b} \mathbb{1}[\hat{y}_i = y_i], \qquad \bar{c}(B_b) = \frac{1}{|B_b|} \sum_{i \in B_b} c_i. \tag{29}$$

ECE is the size-weighted average absolute gap between these quantities:

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|B_b|}{N} \left| \mathrm{acc}(B_b) - \bar{c}(B_b) \right|. \tag{30}$$

Lower ECE is better. Intuitively, if a model reports $c_i \approx 0.9$ on many examples, then roughly 90% of those examples should be correct; large systematic deviations indicate over- or under-confidence. ECE is a coarse summary—it depends on binning and uses only the maximum predicted probability—but it is useful here because it separates confidence alignment from headline discrimination and highlights cases where a method improves macro F1 without improving the reliability of its reported probabilities. Reliability diagrams plot empirical accuracy against mean predicted confidence within each bin and make systematic over- or under-confidence visible as departures from the diagonal.

Post-hoc calibration (optional diagnostic). The main tables intentionally report raw softmax outputs at the final epoch. When full validation and test probability matrices are available, the same held-out protocol can fit post-hoc calibrators on the validation split and evaluate them on the test split without retraining the backbone. Temperature scaling divides logits by a single positive temperature T fit by validation NLL minimization (Guo et al., 2017). Vector scaling applies per-class weights and biases to logits before softmax and is a stronger affine correction in the same family. Dirichlet calibration maps logits through a linear layer into nonnegative concentration parameters and normalizes them to a probability vector (Kull et al., 2019). These methods do not change argmax rankings when T is applied uniformly, but they can substantially lower NLL, Brier score, and ECE when the base scores were miscalibrated. They are especially relevant for CFAL, whose affinity outputs are ranking scores rather than likelihood-based probabilities.

5 Results

Both benchmarks share slide-disjoint splits, the Virchow2 feature family, and a maximum of 30 feature instances per slide, but they answer different prediction questions. The patch benchmark treats each selected crop as an independent labeled example; the WSI-bag benchmark predicts one slide label from an attention-weighted aggregation over the selected instances on that slide. Patch metrics therefore characterize patch-level classification only and must not be read as slide-level diagnostics; WSI-bag metrics characterize slide-level MIL only. Absolute scores and method orderings are not comparable across the two tables.

5.1 Patch-Level Results

The patch benchmark compresses into a narrow performance band once patches are represented with frozen Virchow2 embeddings (Table 4). CFAL ranks highest on both macro F1 (0.807 ± 0.007) and balanced accuracy (0.801 ± 0.003), with paired gains over plain CE of about 0.031 on each headline metric in all three seeds (Table 7). The Scholz-style hybrid objectives remain second: they improve macro F1 by only a few thousandths over CE, whereas CFAL provides a substantially larger shift without balanced oversampling. With the fixed support-tier partition, D&C reaches macro F1 0.746 ± 0.008 and balanced accuracy 0.735 ± 0.004, trailing CE by 0.030 and 0.035 on the paired headline metrics.

Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.858 ± 0.004	0.801 ± 0.003	0.807 ± 0.007
CE + soft MCC (balanced)	0.837 ± 0.007	0.782 ± 0.008	0.782 ± 0.007
CE + soft F1 (balanced)	0.837 ± 0.004	0.782 ± 0.004	0.782 ± 0.004
Weighted CE	0.830 ± 0.004	0.774 ± 0.006	0.776 ± 0.012
CE	0.834 ± 0.002	0.770 ± 0.003	0.776 ± 0.002
Balanced sampler	0.832 ± 0.005	0.777 ± 0.009	0.775 ± 0.008
ProGAN augmentation	0.830 ± 0.003	0.767 ± 0.011	0.774 ± 0.008
Focal	0.830 ± 0.004	0.760 ± 0.002	0.766 ± 0.003
D&C	0.804 ± 0.003	0.735 ± 0.004	0.746 ± 0.008

Table 4: Patch-level test results over three seeds. Methods share the same controlled manifest, frozen Virchow2 encoder, shared MLP trunk, and evaluation code. D&C uses the predefined slide-support tiers for its three expert groups.

The simpler patch interventions separate mainly by what they fail to improve under this fixed protocol. Class-weighted CE is effectively tied with CE on mean macro F1, balanced sampling is slightly lower on macro F1 despite similar balanced accuracy, and ProGAN augmentation stays within the non-generative spread despite adding a large synthetic tail-class volume (Table 5). The accompanying FID and Virchow2 nearest-neighbor diagnostics indicate that this is not merely a quantity problem: many generated patches are visually degenerate or near-monochrome and are distant from the real feature manifold used by the classifier (Figure 4). Focal loss is weakest on macro F1, consistent with the interpretation that down-weighting easy examples is not helpful when most cancer types are already separable in the frozen embedding space.

Class	Real train	Generated	Inception FID	Virchow2 mean NN
Lymphoid Neoplasm Diffuse Large B-cell Lymphoma	600	16,000	315.9	52.16
Cholangiocarcinoma	660	15,940	232.2	49.09
Pheochromocytoma and Paraganglioma	930	15,670	301.3	52.59
Uveal Melanoma	1,160	15,440	255.9	52.46
Rectum adenocarcinoma	1,350	15,250	178.5	51.31
Uterine Carcinosarcoma	1,460	15,140	161.8	49.41
Mesothelioma	1,500	15,100	249.1	46.97
Kidney Chromophobe	1,740	14,860	207.7	40.87
All augmented classes (31 total)	166,480	348,120	178.5 median	47.89 median

Table 5: ProGAN augmentation diagnostics for seed 0. “Generated” counts are synthetic patches added per class to reach the head-class training patch count. Inception FID compares generated patches to the real training subset used for GAN training. Virchow2 mean NN is the mean, over a fixed synthetic subsample, of the minimum L_2 distance to real same-class embeddings in the frozen feature cache.

The classwise-recall plot (Figure 6) shows that remaining errors are distributed across many cancer types rather than concentrated in a single tail class. In this setting, the main practical distinction is therefore not whether imbalance mitigation works at all, but which mechanism best preserves rare-class recall when the feature extractor is already strong. These conclusions remain confined to the patch benchmark: high patch accuracy or macro F1 does not directly establish slide-level utility, because each test prediction is made on a single crop rather than by aggregating evidence across a slide bag.

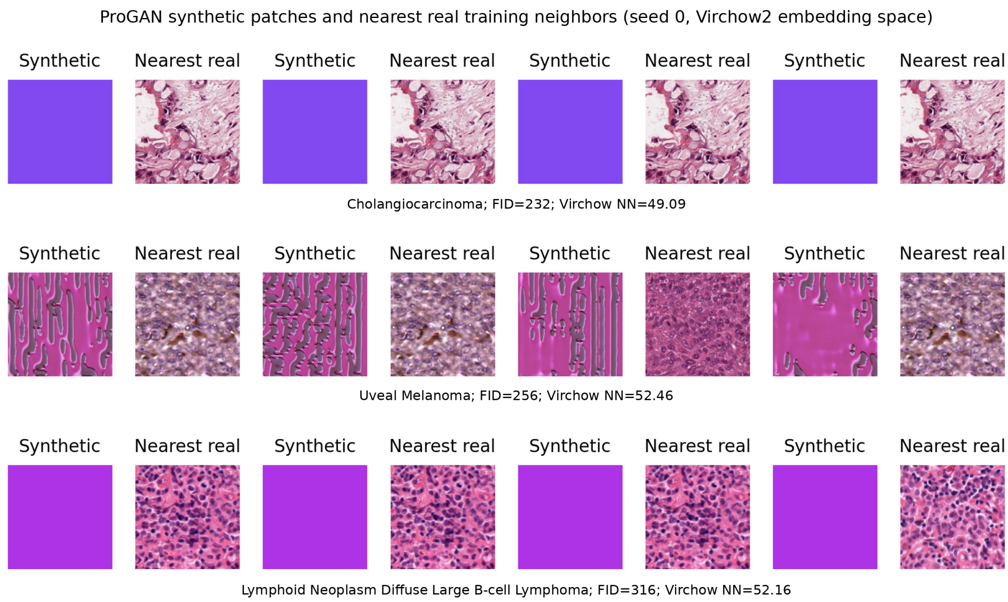


Figure 4: ProGAN synthetic patches (left column of each pair) and their nearest real training neighbors in Virchow2 embedding space (right column) for three tail classes in seed 0. The near-monochrome and visibly unrealistic synthetic patches are diagnostic failure cases rather than plausible histology examples; they explain why ProGAN augmentation is interpreted together with FID, nearest-neighbor distance, and classifier performance.

5.2 WSI-Bag Results

Method	Accuracy	Balanced accuracy	Macro F1
MDE-MIL (ensemble)	0.892 ± 0.020	0.847 ± 0.023	0.849 ± 0.030
RankMix	0.894 ± 0.010	0.863 ± 0.008	0.848 ± 0.017
MIL CE	0.890 ± 0.006	0.836 ± 0.020	0.840 ± 0.020
Balanced MIL	0.885 ± 0.007	0.837 ± 0.015	0.836 ± 0.013
SC-MIL	0.881 ± 0.007	0.837 ± 0.017	0.833 ± 0.016
Weighted MIL	0.880 ± 0.005	0.834 ± 0.018	0.829 ± 0.018
Focal MIL	0.879 ± 0.004	0.824 ± 0.011	0.827 ± 0.009

Table 6: WSI-bag test results over three seeds. Methods share the same capped feature bags, attention-MIL backbone family, and evaluation code.

The WSI-bag benchmark evaluates one slide-level prediction per diagnostic slide by aggregating capped feature bags from the shared split. The 30-instance cap aligns the per-slide evidence budget with the controlled patch manifest and removes the earlier possibility that larger slides benefit simply from contributing more instances. The remaining difference is architectural and supervisory: MIL learns slide-level attention over a set of instances, whereas the patch classifier treats each crop as an independently labeled example. Table 6 should be read as a within-regime slide-level leaderboard, not as confirmation or refutation of the patch-level ranking.

RankMix consistently improves over MIL CE on both headline metrics in all three seeds (Table 7) and has the highest mean balanced accuracy (0.863 ± 0.008) and lowest mean expected calibration error (ECE 0.022 ± 0.003), although macro-F1 gains remain modest and seed-wise

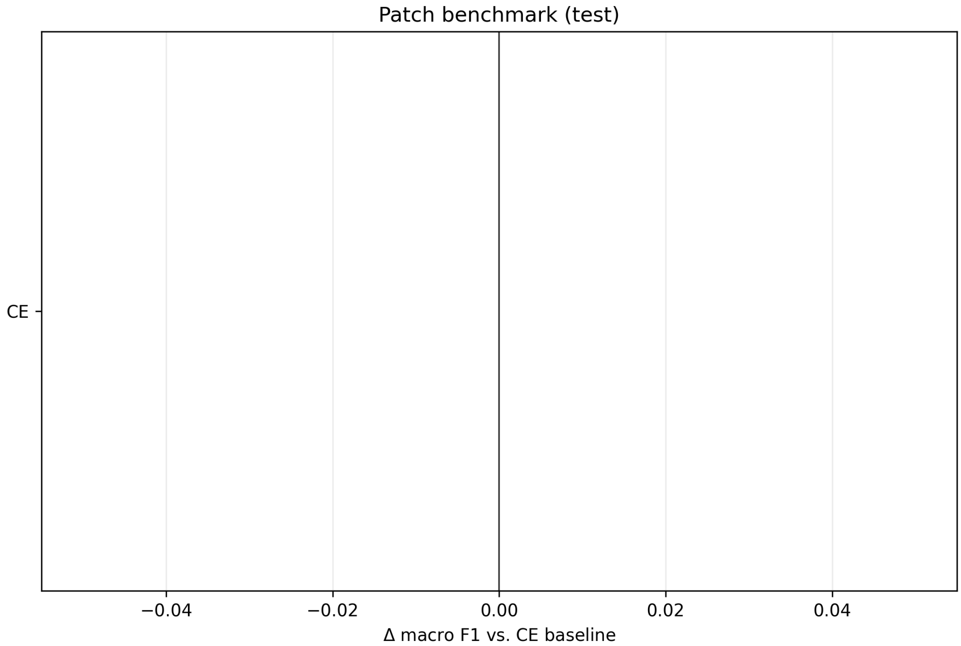


Figure 5: Patch-level change in macro F1 relative to the CE baseline on the held-out test split. Bars show the mean Δ over seeds 0–2 with seed-wise standard deviation. Positive values indicate improvement over CE; near-zero values indicate no meaningful mean change.

standard deviations overlap other methods (Table 6, Figure 9). MDE-MIL has the highest mean test macro F1 (0.849 ± 0.030) but with wider seed spread than RankMix: seed 0 test macro F1 is only 0.819, whereas seeds 1–2 reach 0.879 and 0.850, and its paired macro-F1 delta is negative in seed 0. On mean balanced accuracy and ECE, MDE-MIL sits between RankMix and plain MIL CE; its mean macro F1 is similar to RankMix once seed variability is taken into account.

The remaining WSI methods cluster closely below this pair. Balanced-sampling MIL and SC-MIL produce similar balanced accuracy (0.837), but both trail RankMix and MDE-MIL in macro F1. SC-MIL performs well on validation but does not exceed plain MIL CE on the test macro-F1 average. Weighted MIL CE and focal MIL do not improve over plain MIL CE in this configuration, with focal MIL giving the lowest WSI macro F1. Activation diagnostics confirm that the advanced WSI objectives were active: RankMix generated about 1,100,000 mixed examples per seed, SC-MIL constructed about 1,060,000 positive pairs per seed, and MDE-MIL processed both natural and balanced branches every training step.

5.3 Paired Seed Comparisons

Table 7 makes the headline gains explicit at seed resolution rather than only through aggregate means. CFAL remains above plain CE on macro F1 and balanced accuracy in every seed, with larger shifts than the Scholz-style hybrids. Both patch hybrid objectives also remain above plain CE on macro F1 in every seed, but their aggregate gaps stay within a few thousandths. D&C is below CE in every seed under the fixed support-tier adaptation, with mean paired changes of -0.030 in macro F1 and -0.035 in balanced accuracy. On WSI bags, RankMix remains above plain MIL CE on both metrics in every seed, whereas MDE-MIL improves the mean over MIL CE but with one negative macro-F1 seed (seed 0). The absolute shifts remain modest and would require formal paired testing to treat as significant.

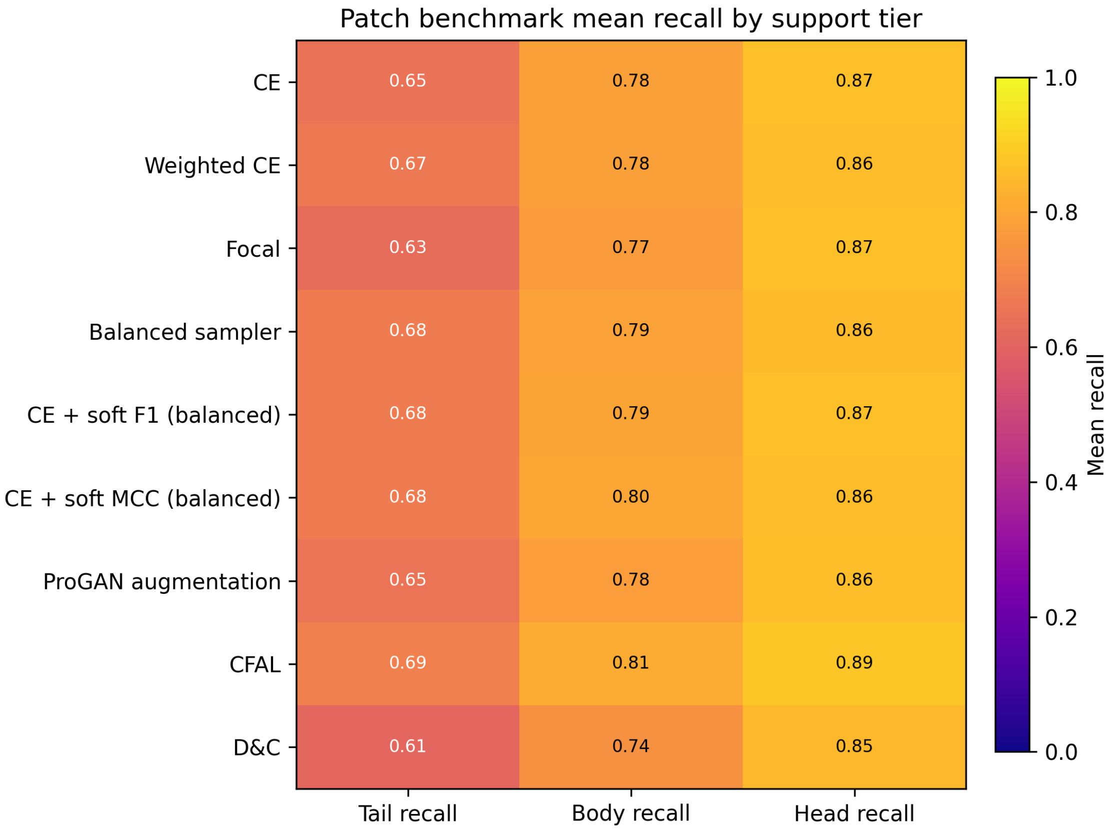


Figure 6: Patch-level classwise recall on the held-out test split. The plot exposes the class-level variation hidden by aggregate accuracy and motivates reporting balanced accuracy and macro F1.

5.4 Validation-Tuned Robustness Check

The fixed-protocol results do not by themselves distinguish a genuinely ineffective mitigation mechanism from an under-tuned one. A targeted robustness check therefore tunes the imbalance-specific knobs of the non-ProGAN mitigation methods on validation macro F1, using balanced accuracy only as a tie-breaker, and then reports the selected configurations on the held-out test split. The tuning grid covers class-weight strength, sampler strength, focal-loss γ, Scholz metric-loss weight, CFAL center-focused exponent γ, D&C cluster count k, RankMix mixing strength, SC-MIL temperature, and MDE-MIL consistency weight λ_{con}; ProGAN is excluded because the generated images show clear quality failures rather than a classifier-only hyperparameter issue.

The tuned robustness check partly narrows the negative result and changes the WSI headline comparison. On patches, validation tuning gives small test gains for weighted CE and balanced sampling, yet CFAL remains the strongest patch method: validation selects the same center-focused exponent $\gamma = 2.0$ as the fixed protocol and reaches test macro F1 0.808 with balanced accuracy 0.800. For D&C, validation selects $k = 10$; its tuned test macro F1 of 0.752 and balanced accuracy of 0.741 remain below CE. The Scholz-style hybrid objectives remain a close second tier, with soft MCC selecting its original metric-loss weight and soft F1 improving only marginally with a larger weight. On WSI bags, validation-selected RankMix and MDE-MIL both reach higher held-out test macro F1 than MIL CE: RankMix with $\alpha = 2.0$ reaches test macro F1 0.856 and balanced accuracy 0.866, whereas MDE-MIL with $\lambda_{\text{con}} = 0.3$—higher than the fixed-protocol default of 0.25—has the highest tuned test macro F1 (0.866) at lower balanced accuracy (0.857). The fixed-protocol MDE-MIL result (0.849) was therefore mildly

Comparison	Δ Macro F1	Δ Balanced accuracy
CE + soft MCC − CE	$+0.006 \pm 0.007$ [+0.001, +0.017, +0.001]	$+0.012 \pm 0.006$ [+0.005, +0.020, +0.011]
CE + soft F1 − CE	$+0.006 \pm 0.005$ [+0.005, +0.012, +0.001]	$+0.012 \pm 0.001$ [+0.013, +0.013, +0.011]
CFAL − CE	$+0.031 \pm 0.007$ [+0.029, +0.041, +0.024]	$+0.031 \pm 0.002$ [+0.029, +0.034, +0.031]
D&C − CE	-0.030 ± 0.008 [−0.038, −0.020, −0.031]	-0.035 ± 0.005 [−0.042, −0.032, −0.032]
RankMix − MIL CE	$+0.008 \pm 0.006$ [+0.017, +0.005, +0.003]	$+0.027 \pm 0.013$ [+0.045, +0.015, +0.020]
MDE-MIL − MIL CE	$+0.010 \pm 0.010$ [−0.004, +0.017, +0.016]	$+0.011 \pm 0.005$ [+0.012, +0.017, +0.005]

Table 7: Paired test-split differences against the within-regime CE baseline on seeds 0–2. Each cell reports mean ± SD across seeds; bracketed values list seed-ordered paired deltas. No formal p-values are computed.

Regime	Method	Selected params	Fixed F1	Val F1	Tuned F1	Bal. acc.
Patch	CE	fixed baseline	0.776	–	0.776	–
Patch	Balanced sampler	sampler_power=0.75	0.775	0.792	0.780	0.774
Patch	CE + soft F1 (balanced)	metric_loss_weight=2	0.782	0.797	0.782	0.780
Patch	CE + soft MCC (balanced)	metric_loss_weight=1	0.782	0.795	0.782	0.782
Patch	CFAL	cfal_gamma=2	0.807	0.819	0.808	0.800
Patch	D&C	dnc_k_clusters=10	0.746	0.766	0.752	0.741
Patch	Focal	focal_gamma=1	0.766	0.781	0.773	0.770
Patch	Weighted CE	weight_power=0.75	0.776	0.793	0.779	0.778
WSI bag	MIL CE	fixed baseline	0.840	–	0.840	–
WSI bag	MDE-MIL (ensemble)	mde_mil_consistency_weight=0.3	0.849	0.869	0.866	0.857
WSI bag	Balanced MIL	sampler_power=0.75	0.836	0.844	0.839	0.842
WSI bag	Focal MIL	focal_gamma=1	0.827	0.839	0.828	0.824
WSI bag	Weighted MIL	weight_power=0.25	0.829	0.848	0.835	0.836
WSI bag	RankMix	rankmix_alpha=2	0.848	0.854	0.856	0.866
WSI bag	SC-MIL	sc_mil_temperature=0.1	0.833	0.848	0.838	0.838

Table 8: Validation-selected tuning robustness check. Configurations are selected only from validation macro F1 averaged over seeds 0–2; held-out test metrics are reported after selection and do not influence the choice. "Fixed F1" and "Tuned F1" are held-out test macro F1; "Bal. acc." is held-out test balanced accuracy.

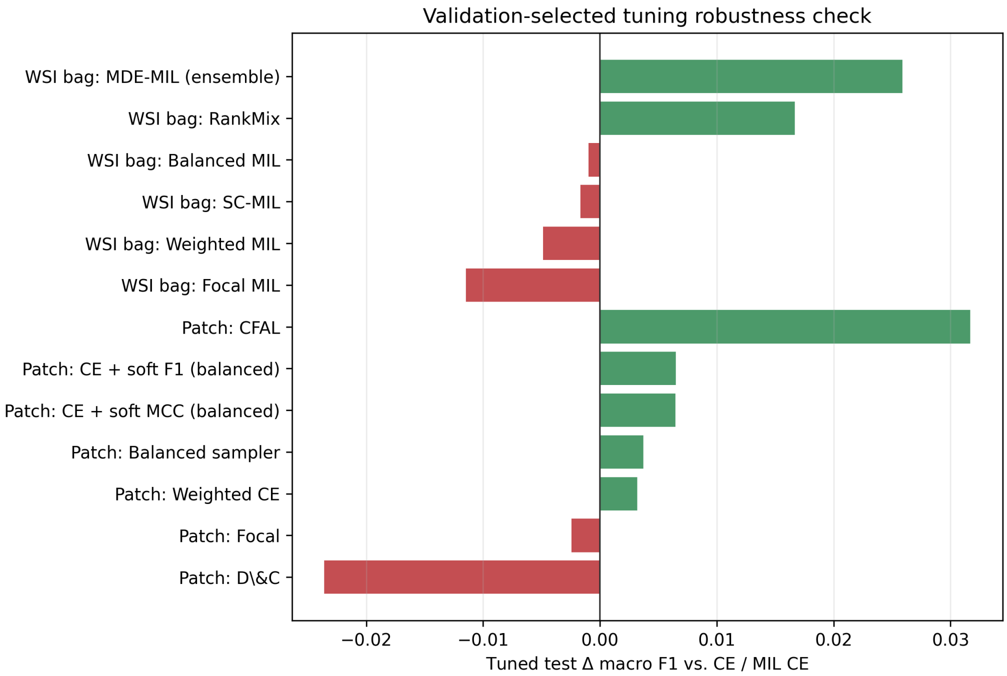


Figure 7: Held-out test macro-F1 change of validation-selected tuned configurations relative to the fixed CE or MIL CE baseline.

under-tuned on consistency weight. Weighted CE, balanced sampling, and SC-MIL improve relative to their fixed settings but still do not surpass MIL CE on test macro F1, and focal loss remains below baseline. After tuning, RankMix has the highest mean balanced accuracy and lowest mean ECE, while MDE-MIL has the highest mean macro F1 (Table 8, Figure 7); these held-out means are not accompanied by formal significance tests.

5.5 Probability Calibration

Table 9 applies the metrics in Section 4.1 and makes the discrimination–calibration tradeoff explicit for raw outputs. On patches, the method with the lowest ECE is not the method with the strongest macro F1, and the Scholz-style hybrid objectives improve class-balanced discrimination without improving likelihood-based calibration. CFAL improves discrimination by the largest margin among patch methods but has the highest raw ECE (0.798 ± 0.003 versus 0.141 ± 0.003 for CE) and the worst NLL/Brier scores in Table 9. The direction matters: CFAL's reported confidences are systematically too high relative to empirical accuracy—a typical over-confidence pattern when geometric affinities are passed through $\log a_{ic}$ and softmax. CFAL should therefore be treated as a strong patch-level *discriminator* under this protocol, not as a calibrated probabilistic classifier on its raw outputs. Conversely, D&C has the lowest patch ECE (0.083 ± 0.003) while producing the lowest patch macro F1. ProGAN augmentation further weakens probability quality relative to the CE baseline. On WSI bags, RankMix is the only method with higher mean balanced accuracy and lower mean ECE than plain MIL CE in this comparison; MDE-MIL lowers mean ECE relative to MIL CE (0.039 versus 0.089) but remains less calibrated than RankMix on mean ECE, while the remaining MIL variants cluster near the baseline.

Table 10 reports the same NLL/Brier/ECE metrics after applying the same post-hoc temperature-calibration protocol to every method. This aligns calibration evaluation with the same best-practice intent used for macro F1 and balanced accuracy comparisons: methods are compared after the required probability mapping is applied consistently rather than only on raw scores.

Regime	Method	NLL	Brier	ECE
Patch	CFAL	2.859 ± 0.016	0.915 ± 0.002	0.798 ± 0.003
Patch	CE + soft MCC (balanced)	1.952 ± 0.092	0.296 ± 0.012	0.140 ± 0.006
Patch	CE + soft F1 (balanced)	2.202 ± 0.089	0.301 ± 0.008	0.144 ± 0.004
Patch	Weighted CE	1.716 ± 0.003	0.302 ± 0.007	0.140 ± 0.003
Patch	CE	1.808 ± 0.087	0.299 ± 0.005	0.141 ± 0.003
Patch	Balanced sampler	1.807 ± 0.105	0.302 ± 0.009	0.141 ± 0.004
Patch	ProGAN augmentation	2.337 ± 0.118	0.314 ± 0.006	0.151 ± 0.003
Patch	Focal	1.156 ± 0.043	0.282 ± 0.006	0.118 ± 0.002
Patch	D&C	0.900 ± 0.022	0.291 ± 0.005	0.083 ± 0.003
WSI bag	MDE-MIL (ensemble)	0.502 ± 0.129	0.170 ± 0.034	0.039 ± 0.015
WSI bag	RankMix	0.407 ± 0.041	0.156 ± 0.015	0.022 ± 0.003
WSI bag	MIL CE	1.034 ± 0.051	0.195 ± 0.009	0.089 ± 0.003
WSI bag	Balanced MIL	1.100 ± 0.050	0.201 ± 0.009	0.091 ± 0.003
WSI bag	SC-MIL	0.958 ± 0.104	0.205 ± 0.013	0.092 ± 0.005
WSI bag	Weighted MIL	1.124 ± 0.028	0.210 ± 0.011	0.096 ± 0.005
WSI bag	Focal MIL	0.763 ± 0.064	0.200 ± 0.007	0.078 ± 0.006

Table 9: Held-out test calibration metrics over three seeds from raw final-epoch probabilities. Lower is better for negative log likelihood (NLL), multiclass Brier score, and expected calibration error (ECE). Values are not comparable across regimes because patch and WSI-bag models operate on different input units.

Regime	Method	NLL	Brier	ECE
Patch	CFAL	0.500 ± 0.020	0.206 ± 0.006	0.015 ± 0.005
Patch	CE + soft MCC (balanced)	0.628 ± 0.025	0.239 ± 0.009	0.042 ± 0.003
Patch	CE + soft F1 (balanced)	0.658 ± 0.019	0.243 ± 0.006	0.054 ± 0.003
Patch	Weighted CE	0.615 ± 0.009	0.244 ± 0.004	0.028 ± 0.002
Patch	CE	0.613 ± 0.019	0.240 ± 0.004	0.037 ± 0.001
Patch	Balanced sampler	0.621 ± 0.026	0.243 ± 0.006	0.033 ± 0.004
Patch	ProGAN augmentation	0.688 ± 0.020	0.254 ± 0.004	0.059 ± 0.010
Patch	Focal	0.593 ± 0.023	0.243 ± 0.006	0.012 ± 0.008
Patch	D&C	0.704 ± 0.013	0.279 ± 0.004	0.029 ± 0.004
WSI bag	MDE-MIL (ensemble)	0.494 ± 0.118	0.171 ± 0.035	0.033 ± 0.012
WSI bag	RankMix	0.393 ± 0.040	0.155 ± 0.015	0.017 ± 0.009
WSI bag	MIL CE	0.409 ± 0.024	0.164 ± 0.008	0.014 ± 0.005
WSI bag	Balanced MIL	0.441 ± 0.016	0.171 ± 0.009	0.013 ± 0.005
WSI bag	SC-MIL	0.437 ± 0.038	0.176 ± 0.013	0.012 ± 0.005
WSI bag	Weighted MIL	0.446 ± 0.014	0.176 ± 0.007	0.012 ± 0.008
WSI bag	Focal MIL	0.441 ± 0.019	0.183 ± 0.007	0.022 ± 0.004

Table 10: Held-out test calibration metrics after method-agnostic post-hoc temperature scaling. For each method and seed, the temperature is fit only on that seed's validation split and then applied to the test split. This table is the decision-facing probability comparison; Table 9 remains the raw-score diagnostic.

On patches, temperature scaling largely removes the raw-score inversion between CFAL and CE: CFAL has the lowest mean NLL (0.500 ± 0.020) and Brier score (0.206 ± 0.006) while reaching mean ECE 0.015 ± 0.005, consistent with treating affinity outputs as rank scores that need a monotone probability map rather than as intrinsically miscalibrated classifiers. Focal loss attains the lowest patch mean ECE (0.012 ± 0.008) after scaling but not the best NLL or Brier; D&C remains weakest on NLL and Brier even after calibration, so its raw ECE advantage does not translate into the best decision-facing probability quality. On WSI bags, RankMix has the lowest mean NLL (0.393 ± 0.040) and Brier score (0.155 ± 0.015); MIL CE improves substantially relative to its raw scores and reaches similar mean ECE (0.014 ± 0.005 versus 0.017 ± 0.009 for RankMix), so the raw RankMix–MIL CE ECE gap should be read as partly a scale mismatch rather than as proof that RankMix is uniformly better calibrated after a shared post-hoc step.

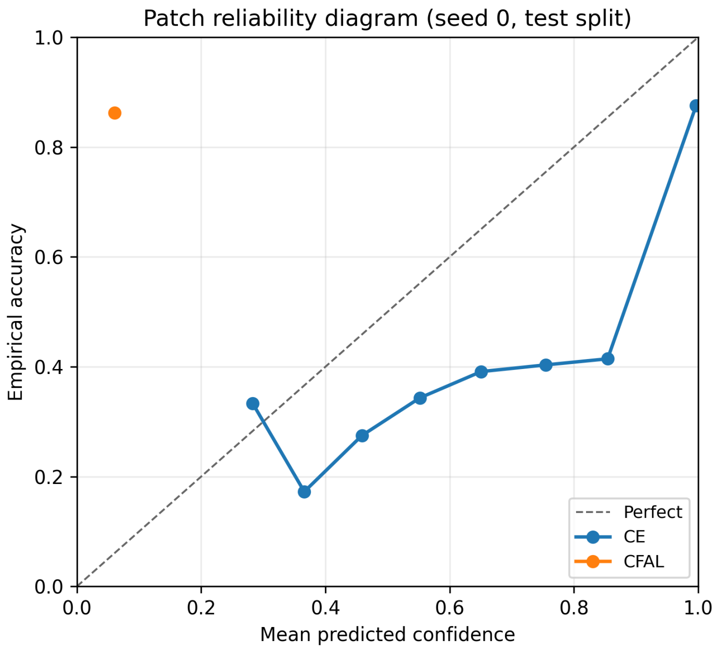


Figure 8: Patch reliability diagram on the held-out test split (seed 0): empirical accuracy versus mean predicted confidence in ten bins. Points below the diagonal indicate over-confidence. CFAL's curve lies far below the diagonal at high confidence, whereas CE tracks the diagonal more closely.

All WSI-bag calibration scores are computed from the same held-out test slides, hard slide labels, and raw softmax outputs of the final-epoch model. An audit over stored test-result files confirms that all seven WSI methods share identical test label sequences within each seed, that reported probabilities sum to one, and that stored NLL, Brier, and ECE values match recomputation from the saved probability matrix. These audits suggest the RankMix–MIL CE calibration gap is not solely an evaluation artifact: plain MIL CE is strongly overconfident (mean maximum predicted probability ≈ 0.98 versus ≈ 0.91 for RankMix), whereas RankMix's soft mixing targets affect training only and not the held-out evaluation forward pass.

5.6 Classwise Difficulty and Support

The classwise recall summaries show that errors are uneven across cancer types, but support tiers alone do not reveal whether rare classes are always the hard classes. A post-hoc difficulty analysis therefore uses the CE baseline in each regime as the reference error pattern and compares each class's full-dataset slide support with its seed-averaged held-out recall. Difficulty is

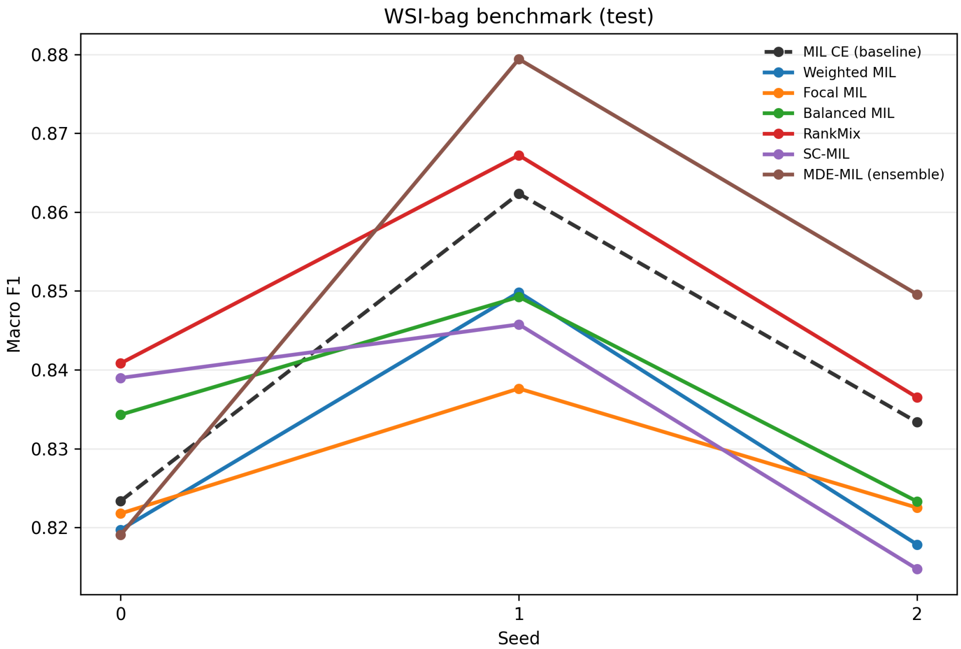


Figure 9: WSI-bag macro F1 on the held-out test split by random seed. The dashed line marks plain MIL CE; seed-wise spread shows why cross-method gaps should be read cautiously when standard deviations overlap (Table 6).

operationalized as the false-negative rate, $1 - \text{recall}$, so the analysis asks whether classes with lower slide support are also the classes with higher baseline error.

Table 11 and Figures 11–12 make the support–difficulty mismatch explicit. Rectum adenocarcinoma is the hardest class in both regimes despite being the fifth-lowest-support class rather than the rarest class. Conversely, the rarest class, diffuse large B-cell lymphoma, does not appear among the five hardest classes in either regime. The rank association between dataset slide support and CE-baseline recall is positive but modest: Spearman $\rho = 0.334$ for patches and $\rho = 0.301$ for WSI bags. These descriptive correlations support a constrained interpretation: class frequency contributes to difficulty, but it does not fully determine which class boundaries remain hard in the frozen Virchow2 feature space.

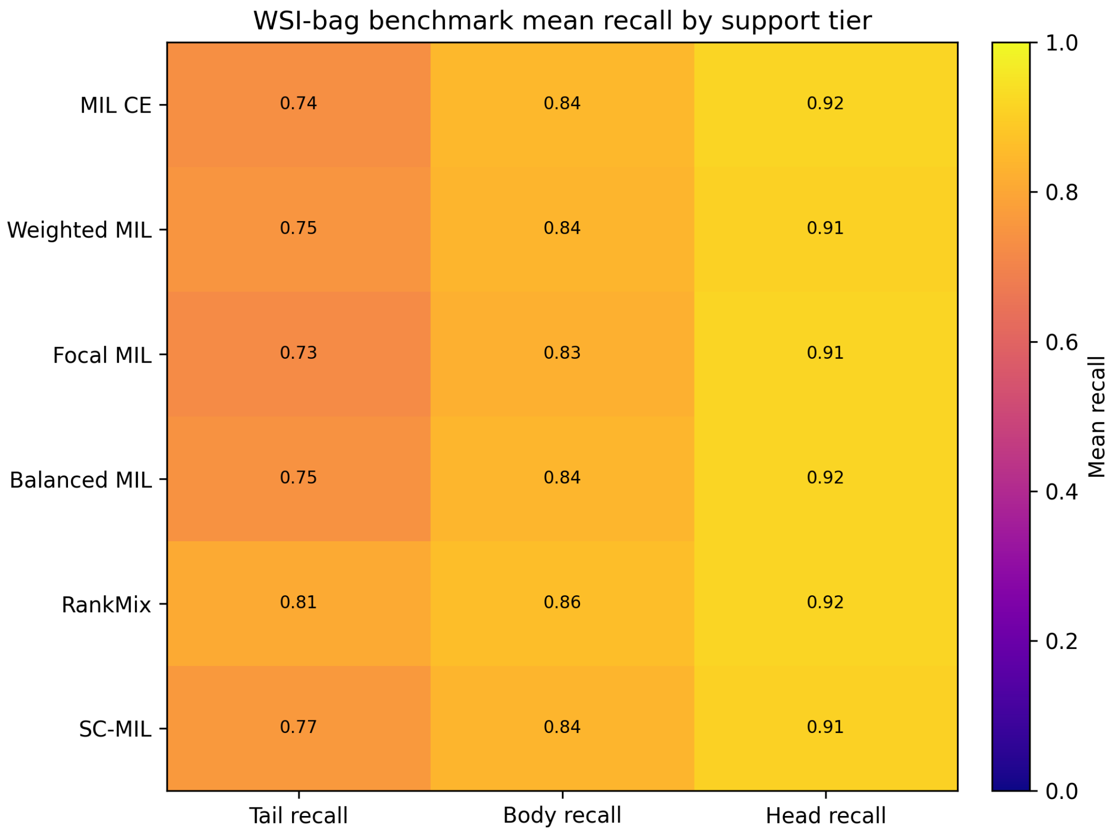


Figure 10: WSI-bag classwise recall on the held-out test split. The spread across classes shows why aggregate accuracy alone is insufficient for the long-tailed setting.

Regime	Class	Slides	Support rank	Difficulty rank	Recall	F1
Patch	Rectum adenocarcinoma	63	5	1	0.111	0.150
Patch	Cholangiocarcinoma	30	2	2	0.342	0.429
Patch	Esophageal carcinoma	113	10	3	0.380	0.411
Patch	Uterine Carcinosarcoma	71	7	4	0.499	0.539
Patch	Cervical squamous cell carcinoma and endocervical adenocarcinoma	209	15	5	0.588	0.619
WSI-bag	Rectum adenocarcinoma	63	5	1	0.348	0.416
WSI-bag	Esophageal carcinoma	113	10	2	0.477	0.506
WSI-bag	Uterine Carcinosarcoma	71	7	3	0.530	0.596
WSI-bag	Cholangiocarcinoma	30	2	4	0.542	0.608
WSI-bag	Cervical squamous cell carcinoma and endocervical adenocarcinoma	209	15	5	0.654	0.720

Table 11: Hardest classes under each regime's CE baseline, averaged over seeds 0–2 on the held-out test split. Support rank is computed from full TCGA-UT slide counts, with rank 1 indicating the lowest-support class. Difficulty rank orders classes by baseline error rate, $1 - \text{recall}$.

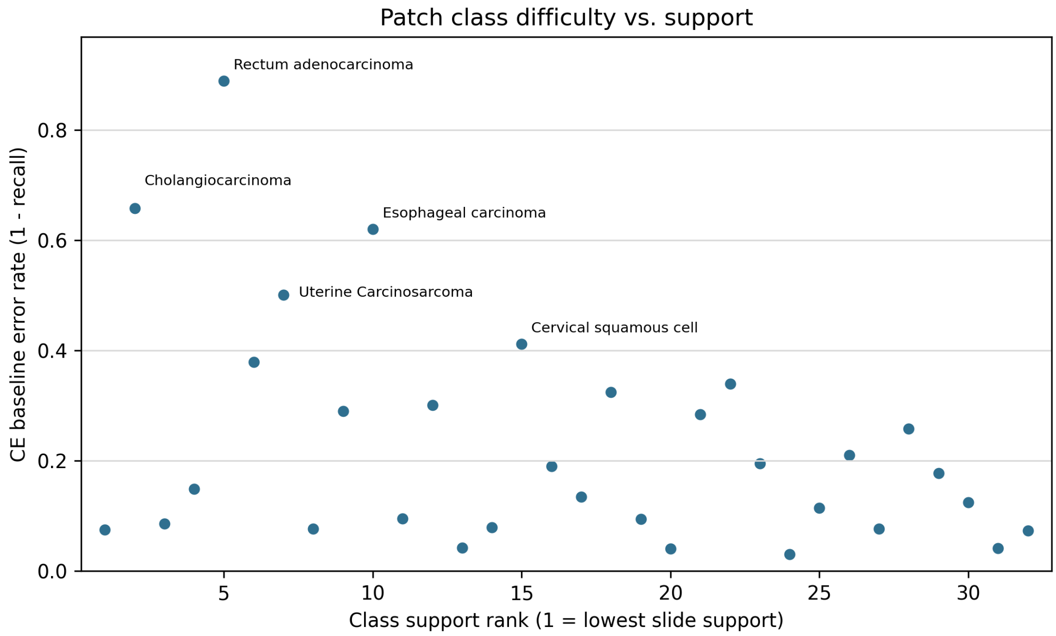


Figure 11: Patch CE-baseline class error rate versus full-dataset class-support rank. The hardest patch classes are not simply ordered by rarity, indicating that exposure-based corrections address only part of the error pattern.

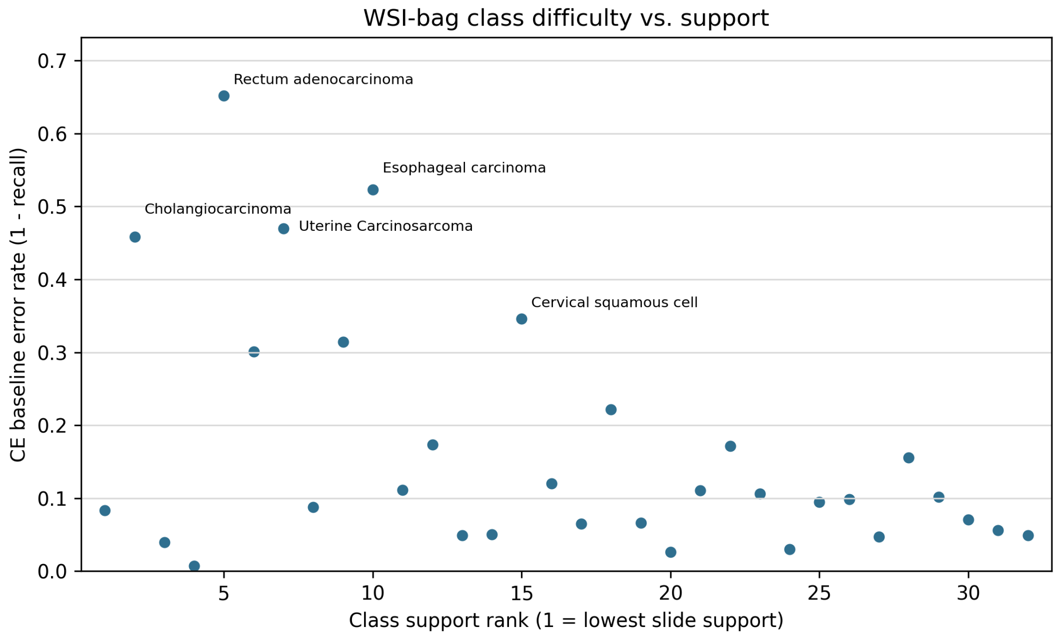


Figure 12: WSI-bag CE-baseline class error rate versus full-dataset class-support rank. The same high-error classes remain visible at slide level, showing that class difficulty is not reducible to class frequency.

6 Discussion

The two benchmarks support different but compatible conclusions. In the patch benchmark, a strong frozen encoder compresses most methods into a narrow band near macro F1 0.77–0.78, but CFAL breaks out of that band (0.807) by replacing the linear head with prototype-based affinity training. D&C's fixed support-tier adaptation falls below CE on both macro F1 and balanced accuracy, despite producing better calibrated probabilities by the reported metrics. The Scholz-style hybrid objectives provide smaller but consistent improvements over plain CE rather than large separations. ProGAN augmentation does not deliver the same advantage in this regime even though it adds large numbers of synthetic tail-class patches; the accompanying quality diagnostics suggest a combination of high Inception FID and substantial Virchow2 embedding-space separation from real training patches, rather than a simple failure to increase tail-class exposure. Focal loss is also unhelpful here, consistent with the interpretation that difficulty reweighting is redundant when the representation has already absorbed much of the class-separation problem. In the WSI-bag benchmark, RankMix consistently improves over MIL CE on balanced accuracy and calibration across seeds and has the highest mean values on those metrics under the fixed protocol, while MDE-MIL—tested without the source paper's multimodal distillation—has the highest mean macro F1 but with higher seed variance and overlapping bands across methods.

These readings should be interpreted within the benchmark's deliberate scope. The study is a controlled comparison rather than a full reproduction of every original paper: hyperparameters follow fixed configuration values and cited defaults rather than method-specific validation tuning, and both regimes use frozen Virchow2 features rather than end-to-end encoder adaptation. Rankings therefore describe how imbalance mechanisms behave when patch and slide classification are already supported by a strong pathology foundation model, not every published implementation under its preferred training recipe. The ProGAN adaptation likewise keeps the progressive-generator mechanism from Ruiz-Casado-style histology augmentation but applies it to multiclass TCGA-UT rather than reproducing the full GAN-family comparison from the source work.

The native long tail also means that class frequency is only one part of the difficulty signal. The classwise difficulty analysis shows this directly: several low-support classes are hard, but the hardest classes are not exactly the rarest classes, and support explains only a modest share of the classwise recall ordering. This distinction helps explain why pure exposure corrections have limited room to improve the patch benchmark. Balanced sampling and class-weighted CE increase the influence of observed minority examples, but they do not add new within-class diversity or change the separability of visually related classes in the Virchow2 feature space. The same reasoning motivates reporting classwise recall, difficulty, and calibration alongside aggregate scores, because a method can improve average class exposure without resolving the hardest class-level decision boundaries.

Holding the encoder fixed within each benchmark therefore changes what an imbalance intervention can accomplish. On patches, the bottleneck shifts from learning stain and tissue morphology to calibrating decision boundaries under a long-tailed label distribution. Under that bottleneck, CFAL's prototype shaping provides the largest fixed-protocol gain, whereas objectives that approximate macro F1 or MCC on balanced minibatches help more modestly. Both families can improve discrimination without improving likelihood scores or ECE relative to CE. On slides, the bottleneck instead includes instance selection and bag aggregation, so RankMix can still add value by mixing informative subsequences even when the underlying features are strong, and in this benchmark it suggests gains on calibration as well as balanced accuracy relative to MIL CE. MDE-MIL shows that coupling natural and balanced branches with cross-expert consistency can raise mean macro F1 without distillation; validation tuning further suggests that a stronger consistency weight ($\lambda_{\text{con}} = 0.3$) lifts held-out test macro F1 to 0.866, although the gain remains seed-sensitive and RankMix still has the highest mean balanced accuracy and lowest mean ECE.

This pattern reinforces the need to evaluate imbalance methods in their native input regimes and at the stage where the model is actually limited. A patch generator should not be judged by first converting synthetic images into one-instance MIL bags, and a MIL feature-mixing method should not be compared directly against a patch-level linear head on single embeddings. The shared split, encoder family, and 30-instance slide budget link the two benchmarks descriptively, but they do not license cross-regime inference: patch accuracy does not directly imply slide-level deployment quality, WSI-bag gains may reflect slide-native aggregation behavior absent from the patch classifier, and neither absolute scores nor method orderings should be transferred across tables. The controlled design narrows the claims but makes them easier to defend: within each benchmark, the split, encoder family, head architecture, input budget, and evaluation code are shared. It also means that weak results should be read as weak results under the chosen fixed hyperparameters, not as proof that the mitigation families cannot work.

Several limitations remain. TCGA-UT patch labels come from pathologist-selected tumor regions; they are useful patch-level class labels but not dense annotations. The two benchmarks share slide-disjoint splits but differ in prediction unit and per-slide evidence budget, so their result tables remain non-transferable as discussed above. The MDE-MIL benchmark omits the CONCH distillation component of the source method and therefore tests the ensemble branch in isolation. CFAL uses L2-normalized Virchow2 embeddings with $\sigma = 1.0$ rather than the AlexNet-scale $\sigma = 130$ reported in the source paper. For D&C, TCGA-UT lacks ROI-level benign labels, so the evaluated adaptation replaces the HiPath partition with fixed slide-support tiers. Main-result tables use fixed, shared training budgets and cited defaults; Section 5.4 adds a targeted validation search over one imbalance-specific knob per non-ProGAN method but does not exhaust architecture, training schedule, or ProGAN generator choices.

7 Conclusion

TCGA-UT supports a cleaner imbalance study when patch-level and WSI-bag supervision are evaluated separately with regime-appropriate inputs. On controlled patch classification with frozen Virchow2 embeddings, CFAL reaches the highest mean macro F1 (0.807), clearly above plain CE (0.776) and Scholz-style CE plus soft F1 or soft MCC with balanced sampling (0.782), whereas ProGAN augmentation and focal loss do not improve over that baseline. On WSI bags, RankMix consistently improves balanced accuracy and calibration across seeds under the fixed protocol, while MDE-MIL has the highest mean macro F1 without multimodal distillation; validation tuning suggests both methods exceed MIL CE on held-out test macro F1, with MDE-MIL highest on that metric ($\lambda_{\text{con}} = 0.3$) and RankMix highest on mean balanced accuracy ($\alpha = 2.0$). These two result sets are complementary rather than mutually confirming: patch-level metrics do not directly establish slide-level utility, WSI-bag metrics reflect slide-level aggregation under the same 30-instance slide budget, and methods should not be ranked across regimes. The practical implication is that imbalance mitigation should be matched to the supervision unit and to the learning stage that still limits performance; otherwise, an intervention can appear ineffective because the representation already absorbs the problem it was designed to solve, or because it was tested in the wrong regime.

A Terminology

Term	Meaning in this study
Slide / WSI	Diagnostic whole-slide image carrying one cancer-type label.
Patch	256×256 H&E crop from a selected tumor region; patch labels inherit the slide cancer type.
Patch embedding	Frozen Virchow2 vector for one controlled patch; input to the patch-level classifier head.
Region	Tumor area selected before patch extraction.
Cancer type	The 32-way class label used for patch and slide-level classification.
Feature chunk	Frozen Virchow2 feature representation used to assemble slide-level feature bags.
WSI bag	Set of patch-feature instances for one slide, consumed by the MIL model.
Head/body/tail class	Fixed support tier from the full TCGA-UT slide distribution; see Section 3.8.
Slide-disjoint split	Split rule preventing any slide's patches or features from crossing train, validation, and test.

Table 12: Terminology used for the two native input regimes (see Section 2.1).

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